Reader: But Figure 13 shows that two **different** numbers in D correspond to one number in E.

Author: It does not contradict the statement "each number in D corresponds to one number in E". I never said that **different** numbers in D must correspond to different numbers in E. Your remark (which actually stems from specific characteristics of the chosen function) is of no principal significance. Several numbers in D may correspond to one number in E. An inverse situation, however, is forbidden. It is not allowed for one number in D to correspond to more than one number in E. I emphasize that each number in D must correspond to **only one** (not more!) number in E.

Now we can formulate a mathematical definition of the numerical function.

Definition: Take two numerical sets D and E in which each element x of D (this is denoted by $x \in D$) is placed in one-to-one correspondence with one element y of E. Then we say that a function $y=f(x)$ is set in the domain D, the range of the function being E. It is said that the argument x of the function y passes through D and the values of y belong to E.

- Sometimes it is mentioned (but more often omitted altogether) that both D and E are subsets of the set of real numbers R (by definition, R is the **real line**).
- On the other hand, the definition of the function can be reformulated using the term "mapping". Let us return again to Fig 13. Assume that the number of arrows from the points of D to the points of E is infinite (just imagine that such arrows have been drawn from each point of D). Would you agree that such a picture brings about an idea that D is **mapped** onto E?

Reader: Really, it looks like mapping.

Author: Indeed, this mapping can be used to define the function.

Definition: A numerical function is a mapping of a numerical set D (which is the domain of the function) onto another numerical set E (the range of this function).

Thus, the numerical function is a mapping of one numerical set onto another numerical set. The term "mapping" should be understood as a kind of numerical correspondence discussed above. In the notation $y=f(x)$, symbol f means the function itself (i.e. the mapping), with $x \in D$ and $y \in E$.